

Parcel Delivery

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities and m routes through which parcels can be carried from one city to another city. For each route, you know the maximum number of parcels and the cost of a single parcel.

You want to send k parcels from Syrjälä to Lehmälä. What is the cheapest way to do that?

Input

The first input line has three integers n , m and k : the number of cities, routes and parcels. The cities are numbered $1, 2, \dots, n$. City 1 is Syrjälä and city n is Lehmälä.

After this, there are m lines that describe the routes. Each line has four integers a , b , r and c : there is a route from city a to city b , at most r parcels can be carried through the route, and the cost of each parcel is c .

Output

Print one integer: the minimum total cost or -1 if there are no solutions.

Constraints

- $2 \leq n \leq 500$
- $1 \leq m \leq 1000$
- $1 \leq k \leq 100$
- $1 \leq a, b \leq n$
- $1 \leq r, c \leq 1000$

Example

Input	Output
4 5 3 1 2 5 100 1 3 10 50 1 4 7 500 2 4 8 350 3 4 2 100	750

Explanation: One parcel is delivered through route $1 \rightarrow 2 \rightarrow 4$ (cost $1 \cdot 450 = 450$) and two parcels are delivered through route $1 \rightarrow 3 \rightarrow 4$ (cost $2 \cdot 150 = 300$).